

Chapter 3

Use of Water Ice for Removal of Surface Contaminants

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1 INTRODUCTION

Alternative nonsolvent cleaning methods are increasingly being sought to replace conventional cleaning solvents, such as chlorinated compounds, hydrochlorofluorocarbons, trichloroethane, and other ozone-depleting solvents. The primary motivation is concern about ozone depletion, global warming, and air pollution, which has led to new regulations and mandates for the reduction in the use of these solvents. Many of these solvents are commonly used for industrial and precision cleaning in a variety of applications, but they have been deemed detrimental to the environment and there is a specific schedule for the phaseout of these solvents [1–3].

Among these alternative technologies are various blasting processes that involve a solid projectile (or cleaning medium) impacting the target surface to be precision cleaned [4]. Conventional blast media, such as sand, plastic chips, glass beads, ceramic powders, and similar materials, generate dust upon impact which must be removed in high-precision cleaning applications. This often entails elaborate removal processes at significant costs, making the removal process technically, economically, and environmentally nonviable. For example, decontamination of radioactive lead bricks using conventional blast media results in generation of hazardous mixed waste stream of lead compounds, radioactive isotopes, and blast media fines that are extremely difficult to dispose. In recent years, there has been increasing interest in the use of water ice as a blasting medium for cleaning and treatment of surfaces. For certain applications, ice blasting¹ provides significant advantages over chemical surface treatment, blasting with sand or other abrasive materials, hydroblasting, and blasting with steam or dry ice (solid CO₂). Ice blasting is a simple, nonabrasive, cleaning process that capitalizes on ordinary tap water, compressed air, and electricity to create an environmentally friendly, cost-effective method for removing surface contaminants without the use of chemicals, abrasive materials, high temperatures, or steam. Ice constitutes an easily available, comparatively easy to fabricate blast medium, which subsequent to transmitting kinetic energy to the surface has the ability to change its physical state to liquid leaving no solid residue after use. The technique can be used for cleaning surfaces, removing paint or stripping contaminants from a surface. It can also be used to remove loose material, blips, and burrs from production metal components after machining, and even softer materials, such as organic polymeric materials, including plastic and rubber components, can be processed. The use of water ice is a smart solution for surface cleaning for organizations concerned with minimizing waste produced during commercial and industrial cleaning applications. Compared to other blast cleaning processes, ice blasting does not accumulate ice particles in the waste and thus reduces secondary waste up to 95%. This chapter is focused on the application of water ice blasting for removal of surface contaminants.

2 SURFACE CONTAMINATION AND SURFACE CLEANLINESS LEVELS

Surface contamination can be in many forms and may be present in a variety of states on the surface. The most common categories of contaminants are as follows:

1. Ice blasting is the term commonly used to refer to surface cleaning using solid water ice particles as the impact medium.

- Particles such as dust, metals, ceramics, glass, and plastics in the submicrometer- to macrosized range
- Thin film or molecular contamination that can be organic (hydrocarbons) or inorganic (acids, gases, bases)
- Cationic (such as Na^+ and K^+) and anionic (such as Cl^- , F^- , SO_3^- , BO_3^{3-} , and PO_4^{3-}) contamination
- Microbial contamination (bacteria, fungi, algae).

Other contaminant categories include metals, toxic and hazardous chemicals, radioactive materials, and biological substances that are identified for surfaces employed in specific industries, such as semiconductor, metals processing, chemical production, nuclear industry, pharmaceutical manufacture, and food processing, handling, and delivery. Common contamination sources can include machining oils and greases, hydraulic and cleaning fluids, adhesives, waxes, human contamination, and particulates. In addition, a whole host of other chemical contaminants from a variety of sources may also soil a surface. Typical cleaning specifications are based on the amount of specific or characteristic contaminant remaining on the surface after it has been cleaned.

Space agencies worldwide specify surface precision cleanliness levels for space hardware by particle size (in the micrometer (μm) size range) and number of particles, as well by film contamination represented by nonvolatile residue (NVR) [5,6]. The cleanliness levels are based on contamination levels established in industry standard IEST-STD-CC1246D for particles from Level 1 to Level 1000 and for NVR from Level AA5 ($10 \text{ ng}/0.1 \text{ m}^2$) to Level J ($25 \text{ mg}/0.1 \text{ m}^2$) as shown in Table 3.1 [7]. The cleanliness levels have been revised or redesignated in revision E of this standard [8]. The maximum allowable number of particles for each particle size range has been rounded in revision E, while the NVR designation levels have been replaced with a single letter R followed by the maximum allowable mass of NVR. For example, former NVR Level J has the new designation R25, Level A/2 is now R5E-1, and Level AA5 is now R1E-5. However, this revision has not yet been officially adopted by the space agencies. Table 3.2 lists the surface cleanliness levels for particle contamination in revision E of IEST-STD-CC1246. Table 3.3 compares NVR contamination designation levels between revisions D and E of the standard.

- Particle fallout specification is important in many space programs for sensitive hardware exposed to cleanroom environments for extended periods. Table 3.4 lists particle fallout counts for different cleanliness levels. The specification method uses an “F” descriptor to designate the fallout requirement. As an example, a specification “*IEST-STD-CC1246 Level 100+F100*” indicates an initial surface cleanliness Level 100 verified per Table 3.2 and Level 100 as the additional particle contamination allowed due to fallout per Table 3.4. In this case, the final cleanliness level of the hardware must meet the particle contamination level shown in column 3 of Table 3.5.
- Visibly clean (VC) is frequently specified for surface cleanliness of space-flight hardware free of all visible contamination, including particles and

NVR. This is a qualitative specification. Several different VC cleanliness levels can be specified based on incident light illumination intensity and the viewing distance from the surface. Table 3.6 defines the VC levels in IEST-STD-CC1246E, which also shows the equivalence with NASA JSC standard [5,9] developed for the manned spaceflight program.

TABLE 3.1 Product Surface Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision D [7]				
Cleanliness Level	Particulate Level		NVR Level	
	Particle Size (μm)	Maximum Allowable Count per 0.1 m ² /or 0.1 L of Gas or Liquid	Level	Quantity (mass/0.1 m ² or mass/0.1 L (gas or liquid))
1	1	1	AA5	10 ng
5	1	2.8	AA4.7	20 ng
	2	2.3	AA4.3	50 ng
	5	1	AA4	100 ng
10	1	8.4	AA3.7	200 ng
	2	6.9	AA3.3	500 ng
	5	2.9	AA3	1 μg
	10	1	AA2.7	2 μg
25	2	53.1	AA2.3	5 μg
	5	22.7	A/100	10 μg
	15	13.3	A/50	20 μg
	25	1	A/20	50 μg
50	5	166	A/10	100 μg
	15	24.6	A/5	200 μg
	25	7.2	A/2	500 μg
	50	1	A	1 mg
100	5	1780	B	2 mg
	15	264	C	3 mg
	25	78.4	D	4 mg
	50	10.7	E	5 mg
	100	1	F	7 mg

Continued

TABLE 3.1 Product Surface Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision D [7]—Cont'd

Cleanliness Level	Particulate Level		NVR Level	
	Particle Size (µm)	Maximum Allowable Count per 0.1 m ² /or 0.1 L of Gas or Liquid	Level	Quantity (mass/0.1 m ² or mass/0.1 L (gas or liquid))
200	15	4180	G	10 mg
	25	1230	H	15 mg
	50	169	J	25 mg
	100	15.8		
	200	1		
300	25	7450		
	50	1020		
	100	95		
	250	2.2		
	300	1		
500	50	11,800		
	100	1090		
	250	26.3		
	500	1		
750	50	95,800		
	100	8910		
	250	213		
	500	8.1		
	750	1		
1000	100	42,600		
	250	1020		
	500	38.7		
	750	4.7		
	1000	1		

TABLE 3.2 Product Surface Particle Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision E [8]

Size Bins		Maximum Particle Count per 0.1 m ² for Each Cleanliness Level								
Min (μm)	Max (μm)	25	50	100	200	300	400	500	750	1000
5	15	19	141	1519						
15	25	2	17	186	2949					
25	50	1	6	67	1069	6433				
50	100	0	1	9	154	926	3583	20,726		
100	250	0	0	1	15 ^a	92	359	1073	9704	
250	500	0	0	0	0	2 ^a	8 ^a	25	205	983
500	750	0	0	0	0	0	0	1	7	33
750	1000	0	0	0	0	0	0	0	1	3
1000	1250	0	0	0	0	0	0	0	0	1

^aUnder no circumstances more than one particle larger than the level designation is allowed per 0.1 m² of surface area.

TABLE 3.3 Comparison of Surface NVR Cleanliness Level Designations for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revisions D and E [7,8]

Designation (Revision D)	Designation (Revision E)	Maximum Allowable NVR Limit (mass/ 0.1 m ² or mass/0.1 L (Gas or Liquid))
AA5	R1E-5	10 ng
AA4.7	R2E-5	20 ng
AA4.3	R5E-5	50 ng
AA4	R1E-4	100 ng
AA3.7	R2E-4	200 ng
AA3.3	R5E-4	500 ng
AA3	R1E-3	1 µg
AA2.7	R2E-3	2 µg
AA2.3	R5E-3	5 µg
A/100	R1E-2	10 µg
A/50	R2E-2	20 µg
A/20	R5E-2	50 µg
A/10	R1E-1	100 µg
A/5	R2E-1	200 µg
A/2	R5E-1	500 µg
A	R1	1 mg
B	R2	2 mg
C	R3	3 mg
D	R4	4 mg
E	R5	5 mg
F	R7	7 mg
G	R10	10 mg
H	R15	15 mg
J	R25	25 mg

IEST-STD-CC1246 Revision E provides two additional categories for specification of surface cleanliness levels.

TABLE 3.5 Total Particle Fallout Cleanliness Levels for Commercial and Noncommercial Applications Defined in IEST-STD-CC1246 Revision E [8]

Size Bins		Maximum Particle Count per 0.1 m ² for Fallout Cleanliness Level		
Min (μm)	Max (μm)	100	F100	100 + F100
5	15	1519	14	1533
15	25	186	5	191
25	50	67	4	71
50	100	9	2	11
100	250	1	1	2

The full description of surface cleanliness level is specified as IEST-STD-CC1246E Level X Y Z. Here X is the numerical particulate cleanliness level from [Table 3.2](#); Y is the NVR cleanliness level from [Table 3.3](#); and Z is an alternative or additional cleanliness level, such as particle fallout from [Table 3.5](#) or VC cleanliness level from [Table 3.6](#). Some examples are shown below.

X	Y	Z	Full description
300			IEST-STD-CC1246E Level 300 Specifies: particles only
100+F100			IEST-STD-CC1246E Level 100+F100 Specifies: particles and accumulated fallout
200	R2		IEST-STD-CC1246E Level 200 R2 Specifies: particles and NVR
		VC-0.5-1000UV	IEST-STD-CC1246E Level VC-0.5-1000UV Specifies: visibly clean using UV light
		VC-3-500	IEST-STD-CC1246E Level 300 Specifies: visibly clean only
	R5E-1		IEST-STD-CC1246E Level 300 Specifies: NVR only

In several industries, the precision cleanliness level is defined as an organic contaminant level of less than 10 μg of contaminant per cm², although many

TABLE 3.6 Criteria for Different VC Cleanliness Levels in IEST-STD-CC1246E and the Equivalence to the NASA JSC Standard

IEST-STD-CC1246E			NASA JSC Standard		
<i>Cleanliness Level</i>	<i>Viewing Distance^a</i>	<i>Light Intensity^a</i>	<i>Cleanliness Level</i>	<i>Viewing Distance^a</i>	<i>Light Intensity^a</i>
VC-0.5-1000	0.25-0.5 m (~9.84-19.68 in.)	1000 lumens/m ² (~93-ft candles)	Visibly clean highly sensitive (VC-HS)	6-18 in. (~0.15-0.46 m)	100-ft candles (~1076 lumens/m ²)
VC-1-500	0.5-1.0 m (~1.64-3.28 ft)	500 lumens/m ² (~46-ft candles)	Visibly clean Sensitive (VC-S) ^a	2-4 ft (~0.61-1.22 m)	50-ft candles (~538 lumens/m ²)
VC-2-500	1.0-2.0 m (~3.28-6.56 ft)	500 lumens/m ² (~46-ft candles)	No equivalence	-	-
VC-3-500	1.5-3.0 m (~4.92-9.84 ft)	500 lumens/m ² (~46-ft candles)	Visibly clean standard (VC) ^a	5-10 ft (~1.52-3.04 m)	50-ft candles (~538 lumens/m ²)

^aThe viewing distance and light intensity (white light) in the NASA JSC standard are given in U.S. customary units (feet and inches; foot-candles) and are given in metric units in IEST-STD-CC1246E. The converted values are listed in parentheses below the original units for each standard.

applications are setting the requirement at less $1 \mu\text{g}/\text{cm}^2$ [7,8]. These cleanliness levels are either very desirable or are required by the function of parts such as machined parts, electronic assemblies, optical and laser components, precision mechanical parts, and computer parts.

Airborne molecular contamination is chemical contamination in the form of vapors or aerosols that can be organic or inorganic, and it includes everything from acids and bases to organometallic compounds and dopants. Many of the products and manufacturing processes are also sensitive to, or they can even be destroyed by, airborne molecular contaminants (AMCs) that are present due to external, process, or otherwise generated sources, making it essential to monitor and control AMCs. A new standard ISO 14644-10, “Cleanrooms and Associated Controlled Environments—Part 10: Classification of Surface Cleanliness by Chemical Concentration” [10], is now available as an international standard that defines the classification system for cleanliness of surfaces in cleanrooms with regard to the presence of AMCs. In fact, the standard identifies at least eight categories of AMCs—acid (ac), base (ba), biotoxic (bt), condensable (cd), corrosive (cr), dopant (dp), total organic compounds (toc), and oxidant (ox), as well as individual substances or groups of substances. This standard is applicable to all solid surfaces in cleanrooms and associated controlled environments such as walls, ceilings, floors, working environment, tools, equipment, and devices.

For the purpose of this international standard, consideration is only given to the chemical characteristics of a particle. The physical properties of the particle are not considered, and this standard does not cover the interaction between the contamination and the surface. This standard does not include the contamination generation process and any time-dependent influences (deposition, sedimentation, aging, etc.) or process-dependent activities such as transportation and handling. Nor does it include guidance on statistical quality control techniques to ensure compliance with the standard.

The surface cleanliness by chemical concentration (SCC) class is designated by the classification number in Table 3.7 and expressed as the maximum allowable concentration of the specified chemical substance or group of substances in gravimetric (g/m^2) or numeric (number of atoms, ions, or molecules per unit area) units. Very low concentrations of contaminants are often measured in numeric units. These are converted to gravimetric units (g/m^2) by the following equation:

$$C_{\text{SCC}} = \frac{M(C_{\text{SCC-N}})}{N_z}$$

where C_{SCC} is the surface mass concentration; M is the molar mass of the atomic, ionic, or molecular species; $C_{\text{SCC-N}}$ is the surface number concentration; and N_z is Avogadro’s number.

The SCC class number is only valid with a descriptor that includes the chemical substance or group of substances to which the class number is applied and is

TABLE 3.7 Surface Cleanliness by Chemical Concentration According to ISO Standard ISO 14644-10 [10]

Classification Number	Concentration (g/m ²)	Concentration (ng/m ²)	Concentration (ng/cm ²)
0	10 ⁰	10 ⁹	10 ⁵
1	10 ⁻¹	10 ⁸	10 ⁴
2	10 ⁻²	10 ⁷	10 ³
3	10 ⁻³	10 ⁶	10 ²
4	10 ⁻⁴	10 ⁵	10 ¹
5	10 ⁻⁵	10 ⁴	10 ⁰
6	10 ⁻⁶	10 ³	10 ⁻¹
7	10 ⁻⁷	10 ²	10 ⁻²
8	10 ⁻⁸	10 ¹	10 ⁻³
9	10 ⁻⁹	10 ⁰	10 ⁻⁴
10	10 ⁻¹⁰	10 ⁻¹	10 ⁻⁵
-11	10 ⁻¹¹	10 ⁻²	10 ⁻⁶
-12	10 ⁻¹²	10 ⁻³	10 ⁻⁷

expressed in the format ISO-SCC Class N (X), where *N* is the class number and *X* is the chemical substance or group of substances or one of the categories of contaminants mentioned earlier.

3 THEORETICAL CONSIDERATIONS

3.1 Phase Behavior

Ice is a phase change material, changing from solid (below 273 K) to liquid water and evaporating to gas (above 373 K). Ice can exist in multiple crystalline forms depending on the pressure. As shown in the phase diagram for water (Fig. 3.1), the known ices can be divided into the low-pressure regimes (hexagonal ice, cubic ice, and ice XI), high-pressure regimes (ices VII, VIII, and X), and moderate-pressure regimes (200-2000 MPa; ices Ih, II, III, V, and VI) [11 and references therein]. Of these solid phases, nine phases are thermodynamically stable, while six phases are metastable. Virtually, all ice in nature is present as the hexagonal form, ice Ih. A metastable cubic form of ice, ice Ic,

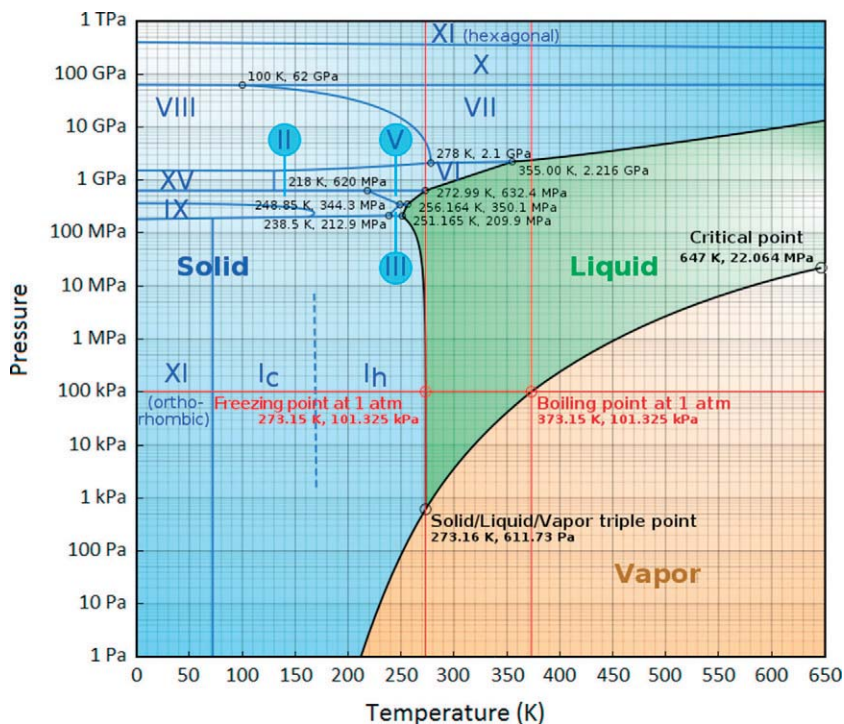


FIGURE 3.1 Phase diagram of water.

has been found in the upper atmosphere, but this transforms to ice Ih at about 183 K. The phase diagram also has a number of triple points and one or more critical points. The triple point of water, when ice, water, and water vapor can coexist, is at a temperature of 273.16 K and a pressure of 611.3 Pa. Thermodynamic models of water and ice and the thermodynamic functions of the phase transitions have been described [12–16].

For the formation of ice for blasting, the phase diagram provides the necessary pressure and temperature conditions. These conditions are fulfilled by commercial blasting systems in which stable ice crystals are continuously generated.

3.2 Mechanical Properties

Contaminant removal is the result of impact of ice particles on the surface. An ideal blast medium should have an edged form with a hardness of at least 6 Mohs, and it should completely transform to vapor at room temperature [17]. The impact behavior of ice particles is determined by its elasticity. In the temperature range 270–233 K, ice behaves as a nearly perfect elastic body. Table 3.8 compares the dynamic elastic properties of ice at 268 K with

TABLE 3.8 Dynamic Elastic Properties of Ice Compared with Conventional Abrasive Materials [18–29]				
Material	Young’s Modulus (GPa)	Shear Modulus (GPa)	Bulk Modulus (GPa)	Poisson’s Ratio
Ice	8.9-9.9	3.4-3.8	7.8-11.8	0.31-0.36
Glass	50-90	19-26	28-66	0.2-0.27
Various aluminum alloys	69-70	24-28	64-78	0.32-0.35

conventional abrasive powder materials such as glass and various aluminum alloys [18–29]. Clearly, the properties of these abrasive materials are superior to those of ice, and they are more effective as abrasives. However, water ice is more acceptable in applications where the surface contaminants are the primary concern and abrasion of the surface cannot be tolerated.

The hardness of ice is an even more decisive property that influences its cleaning performance. As shown in Figs. 3.2 and 3.3, the hardness varies inversely with the temperature of the ice, ranging from Mohs hardness of 2 at a temperature of 263 K, similar to dry ice (solid CO₂) or sulfur or gypsum, and increasing to Mohs hardness of 6-7 at a temperature of 193 K, comparable

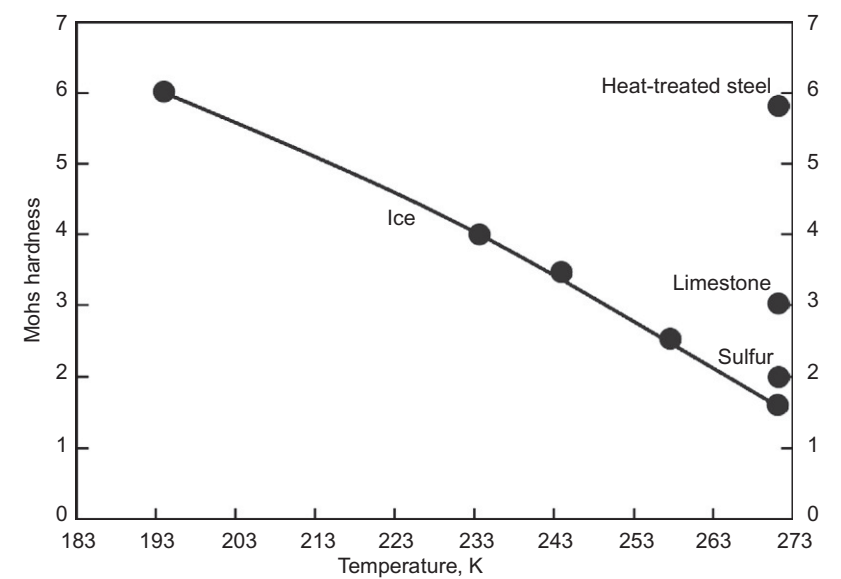


FIGURE 3.2 Mohs hardness of ice as a function of temperature [30–32, 45].

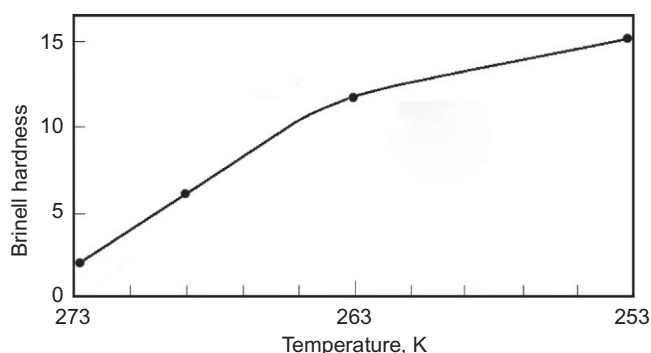


FIGURE 3.3 Brinell hardness of ice as a function of temperature [34, 36, 38–43].

to the hardness of silica glass or hardened steel [30–45]. This property of ice is often not considered or utilized in ice blasting procedures, yet it provides an opportunity for innovative applications of ice blasting, such as deburring metallic materials which requires high hardness. Even though ice does not completely fulfill all three of the requirements for an ideal blast medium since it will melt to a liquid and not transform to vapor, so it is very close to the definition of a perfect blast medium. The temperature dependence of the hardness of ice (Figs. 3.2 and 3.3) makes it the only blast medium that can be adapted to the conditions and requirements of the process, from hard to very soft for application to delicate and highly sensitive substrates.

3.3 Mechanism of Ice Blasting

The primary mechanism of contaminant removal by water ice is based on the impact of ice particles on the surface of the part. The solid ice particles displace surface contaminants by imparting energy from the impact and from lateral deformation of the ice particles. The ice melts after impact and then flushes away the debris. This “scrub and flush” process makes ice blasting effective for removal of surface contaminants. Figure 3.4 illustrates the removal mechanism.

Displacement of the contaminants from the surface occurs when the momentum of the solid ice particle (before phase change) exceeds the inertia of the contaminant. The displacement force is proportional to the cosine of the angle of impact and the square of the radius of the ice particle. Thus, for contamination deposited on a planar surface, maximum displacement force is achieved at 0° [46]. However, surfaces have irregularities, and the optimal displacement is achieved at some nonzero angle (Fig. 3.4a). Also, maximum displacement force is obtained with larger ice particles.

During phase change (Fig. 3.4b), ice particles exert pressure against the surface as these deform, providing a strong ice-surface frictional interaction whereby minute amounts of contaminants can be scrubbed away. The frictional

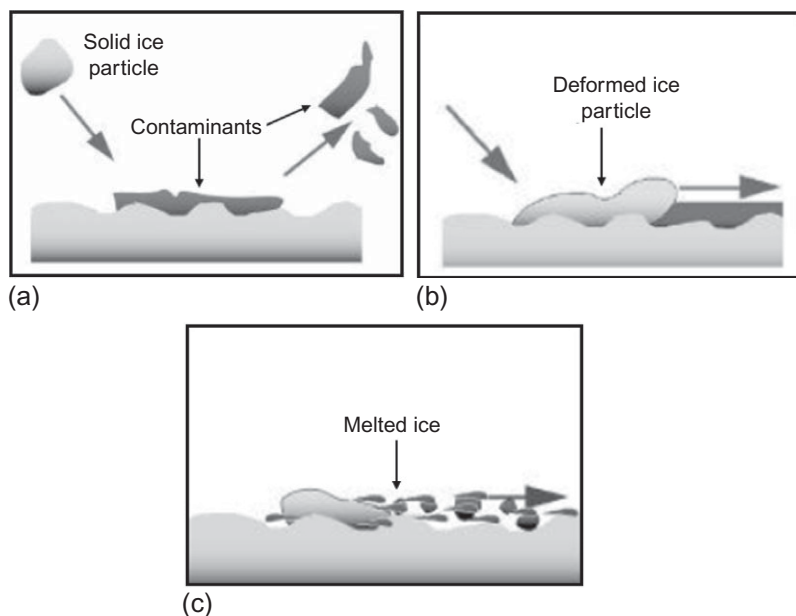


FIGURE 3.4 Mechanism of removal of contaminants by ice particles (a) before, (b) during, and (c) after phase change.

scrubbing pressure is proportional to the sine of the impact angle, so that the maximum force is achieved at 90° from the maximum angle required for displacement [46]. In practice, optimal cleaning can be achieved at between 30° and 60° angle of impact. The scrubbing force is not dependent on the size or mass of the ice particles, but it is substantial. At a blasting pressure of 1 MPa, the frictional scrubbing force is approximately 30 MPa [47].

After phase change, ice particles melt into water to rinse away the removed contaminants (Fig. 3.4c). The fact that water is generated is important for cleaning applications. Water is a solvent that removes all soluble salts. The blast mist also encapsulates the blast debris to control airborne dispersions. This is particularly important for worker safety in the removal of asbestos fibers and radioactive contamination. For very high levels of cleanliness, oil-free compressed air and high-purity deionized water should be used.

In the case of deburring, the kinetic energy of the ice particle is transformed into cutting energy through deceleration [48–50]. At a given impact angle, pressure surges produced by the transmitted kinetic energy of the ice particle result in plastic deformation at the burr. At the same time, additional pressure is generated due to the liquid water from the melting ice particles. When internal pressure builds up to the point where it exceeds the material strength, cracks begin to appear. Spreading and aggregation of these cracks will lead to the removal of surface material [51]. The material strength is also degraded by

low-temperature embrittlement in a wide variety of materials like steels, non-ferrous metals, and plastics. The extremely low temperatures down to 153 K by the utilization of deep frozen ice as blasting abrasive can influence the brittle fracture characteristics of these materials and burrs can be removed more easily. Additionally, thermal shock to the material is influenced by process parameters such as feed rate, nozzle offset, and compressor pressure.

4 DESCRIPTION OF THE PROCESS

The ice blasting process has been employed in a variety of applications for cleaning and deburring operations. Crystalline ice particles are produced continuously at a rate of 90 kg/h. Using a two-hose system, ice particles are transported through a low-pressure hose to the blasting nozzle where a second higher pressure hose delivers up to 1.2 MPa pressure to accelerate the ice particles toward the target surface via different carrier fluids. A typical process flow with water as the carrier fluid is shown in Fig. 3.5 [52]. The high-pressure ice-water

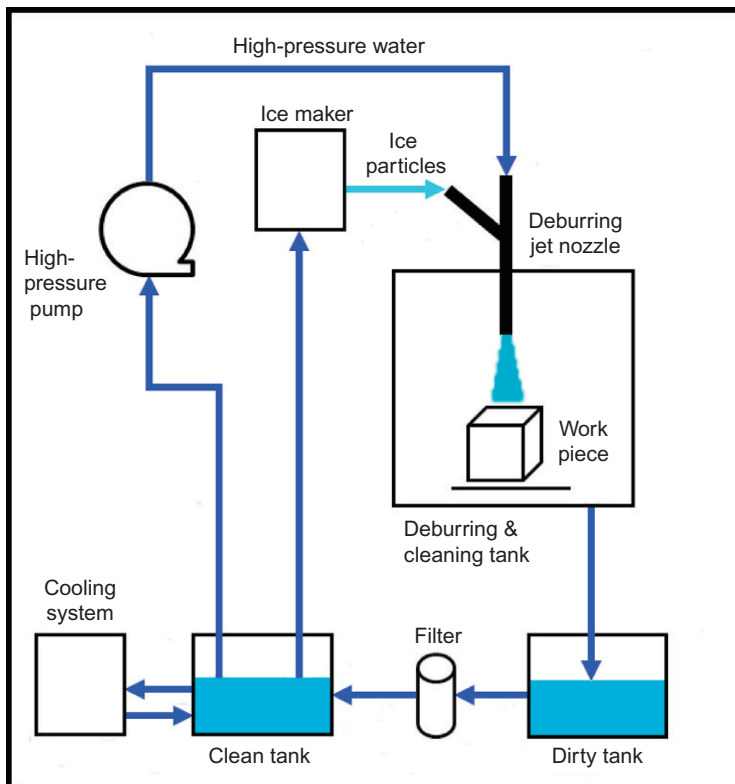


FIGURE 3.5 Typical process flow for metal deburring and cleaning by ice-water jet [52].

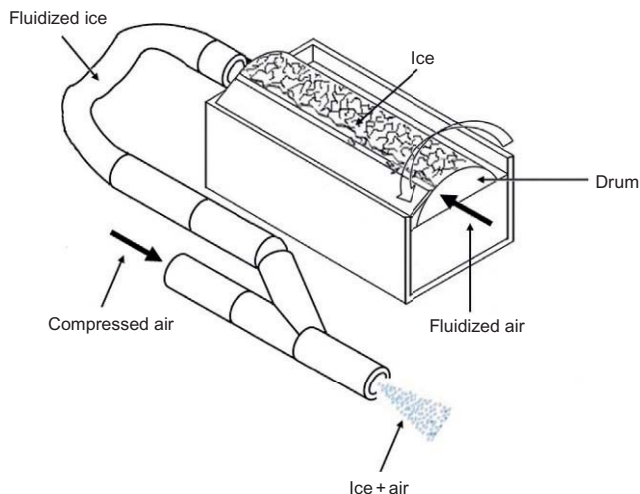


FIGURE 3.6 Typical commercial system for ice blasting using an ice-air jet.

jet impinges on the part to remove the burrs. The ice melts to water which is filtered, cleaned, and recirculated to the ice maker and to the high-pressure pump to form the ice-water jet for processing of new parts. The jet nozzle with multiple axis control enables accurate and precise processing of complex shapes and parts with tight spaces.

Figure 3.6 shows a commercial ice blasting system that uses air as the carrier fluid. Upon impact, the ice particles undergo a phase change, turning approximately half of their solid mass into vapor and the other half into liquid. The amount of liquid residue for collection will vary with the relative humidity and evaporation rate in the ambient atmosphere.

Ice blasting does not generate dust that is common in most abrasive media blasting operations. This is especially important for operators involved in lead-based paint or asbestos abatement jobs where airborne particulate levels must remain low. Ice blasting is nonabrasive and will not leave a profile on most substrates.

Typical water blasting operations produce 300-2000 L of liquid that generally needs to be contained and collected. At 76-90 L/h, a large cost-reduction factor in containment and disposal, especially of hazardous materials, can be realized with ice blasting.

5 CLEANING SYSTEMS

Ice blasting has generated significant commercial interest since the first U.S. patent for producing ice particles and a method for cleaning and polishing automobiles using soft, discrete ice particles [53,54]. This has led to the development of a variety of technologies designed to deliver a high-pressure spray

containing ice particles for performing particular surface treatment procedures [47,52,55–88]. An ice blasting system consists of refrigeration, an icemaker, and a blast nozzle. These are proven commercially available components. Ice blasting typically uses about 90–100 L/h of ordinary tap water to continuously produce crystalline ice particles. For high-precision cleaning, it is essential to use pure water and to maintain high degree of cleanliness of the system.

One reliable ice production method is the immersed cold drum (Fig. 3.7), in which water is continuously frozen on the curved surface of the rotating drum into a thin, curved sheet of ice that is fragmented to form ice particulates [83]. Under appropriate conditions of drum diameter, temperature, and rotational speed, the ice sheet can be formed with sufficient internal stress that when its frontal edge impacts a doctor blade, the sheet fractures into small ice fragments. The fragmented ice particulates are continuously separated from the curved surface by a venturi-type nozzle that draws the ice fragments by vacuum. The particles are directly injected into a stream of air of sufficient velocity to fluidize the particles which flow with no interference. The natural fragmentation of a stressed ice sheet creates the particles without any mechanical means to size them. The fluidized ice particles are ejected at a controlled velocity from the nozzle. Other carrier fluids that have been proposed include nitrogen, high-pressure water, gaseous CO_2 , and even supercritical CO_2 . In a novel innovation, a combination of water ice and dry ice (solid CO_2), propelled by air, is used as the blasting medium [89]. This has the advantage of overcoming the drawbacks of using water ice or dry ice alone as the blasting medium.

Ice particles are stable but can exist only at subzero temperatures. Maintaining such temperatures throughout the delivery system is a difficult task, particularly since the average force of adhesion between the particles increases exponentially from about 10^{-4} N at 258 K to 7×10^{-3} N at 273 K [20,26,90–93]. Ice particles

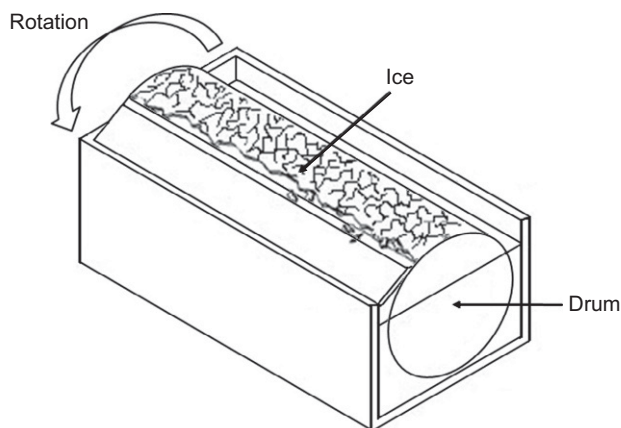


FIGURE 3.7 Immersed cold drum for ice making.

are also not free flowing, and under these thermal conditions, the ice tends to pack and agglomerate when static, thereby clogging the lines. Thus, the supply lines must be maintained at low temperature below 200 K for efficient operation of the system and to achieve optimum physical characteristics of the ice particles for blasting. For the system to work reliably, ice particles must be created and consumed continuously in a dynamic state.

An alternative “fog” process for generating ice particles has been demonstrated [49,50]. As shown in Fig. 3.8, the ice particles are formed by a fog at cryogenic temperatures around 173 K which gives the ice greater hardness and abrasiveness in blasting. The differences between the fog process and the process used in commercially available equipment [47,52,94,95] are primarily in the method of ice particle production with controlled particle parameters such as defined size, form and hardness, and secondarily in the working process and particle temperature in commercial ice generating equipment. Figure 3.9 shows the high-quality uniform ice crystals obtained with the fog process. The commercial processes operate with mechanically scraped ice particles and are not in a cryogenic temperature range. The ice particles have an undefined form and size and are softer and much less abrasive than ice generated in conventional commercial equipment.

To operate an ice blasting machine, only standard utilities such as water, air, and electricity are required. Ice can be formed within 70–90 s after starting the machine and is available for processing. A typical ice blast machine uses 100 L/h of water, 15 kW of electricity, and about 8 m³/min of compressed air. As ice particles disintegrate on impact, they create a blast mist that helps to suppress dust from the operation and to cool the environment for the workers. This is

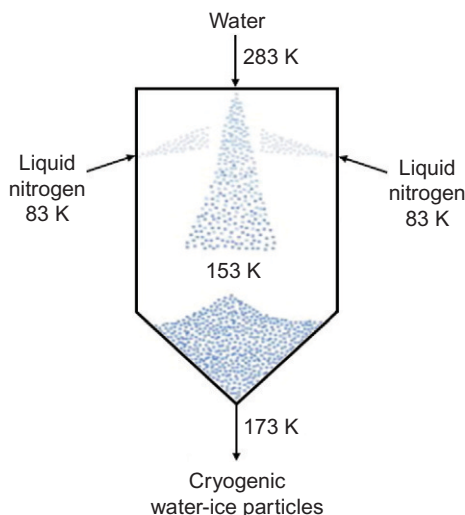


FIGURE 3.8 Production of cryogenic ice crystals by the fog process.

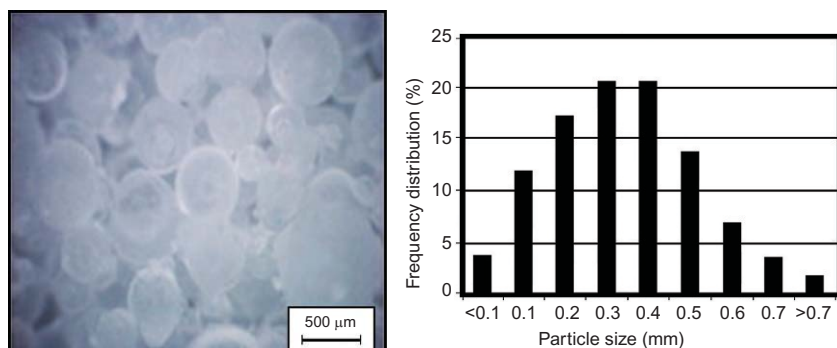


FIGURE 3.9 Wet-ice particles obtained by the fog process in a diameter range of 0.1–0.7 mm with frequency distribution [50].

particularly helpful in the summer. Evaporation will normally reduce the net liquid waste to about 10–40 L/h.

A source for water, air, and electricity is needed to operate the ice blasting machine. Standard water supply is normally found in all plants and facilities. Ice blasting medium is generated from the water by simply connecting the water line to machine. Water is available 24 h a day and is very inexpensive compared to purchasing blast medium of any type. Normal air supply is 175 L/s at 0.8–1.0 MPa; however, blast pressures up to 1.2 MPa can be used for faster cleaning rates on stubborn contaminants and coatings. Depending on the operating environment, an after-cooler may be needed to keep the compressed air supply at temperatures below 308 K. A water separator is recommended in excessively damp or humid operating environments and/or an air filter to reduce excessive compressor blow by. Ice blasting systems require three-phase power, typically 400 V, 50 cycle.

To obtain the best results, a maximum length of 70 m of hose is recommended. This offers a 140 m radius around the machine. Additionally, a maximum vertical height of 20 m is recommended. For portable models, the machine should be moved closer to the work area to stay within the 70 m recommendation.

Figure 3.10 shows a commercial ice blasting and cleaning system [52].

Like all blasting type operations, worker safety should not be compromised. Manual cleaning systems can generate up to 115 dB sound at the nozzle when the higher velocity air meets the slower velocity ice particles. Eye and ear protection is mandatory during blasting, and respiratory protection is recommended. Protective clothing can range from simple Tyvek® suits to keep general debris off the worker to full chemical gear when cleaning hazardous materials. Generally, light rain gear is used for most blasting operations. As with all blasting operations, the blast nozzle should never be pointed at anyone or serious injury can occur.



FIGURE 3.10 Commercial ice blasting unit for cleaning automobile transmission valve bodies. (Courtesy of Advanced Engineering Company, Niigata, Japan.)

5.1 Abrasive Medium Addition

The effectiveness of ice blasting can be enhanced by mixing a small amount of abrasive media with the ice particles. In one commercial system, only about 7 kg of abrasive medium per hour is added to the ice particles [47]. With this mixture, the system has virtually the same cleaning performance as sandblasting and can be used to remove virtually any coating and contamination from almost any surface, including paint, rust, grease, oil, soot, and other contaminants from carbon steel, stainless steel, aluminum, galvanized metal, concrete, ceramic tile, glass, plastics, rubber, and other substrates.

This abrasive on-demand system comprises a small container for the abrasive medium, a pressure valve, and a small hose to feed the medium toward the nozzle (Fig. 3.11). A switch is mounted on the nozzle to open or close the pressure valve. This makes it possible to instantly blast with or without abrasive medium.

6 COST CONSIDERATIONS

A fully automated ice blasting system can cost between US\$450,000 and US\$500,000. Such a system has the capability to process a part every 20 s. Manual systems can start at US\$50,000, but they handle fewer parts. Due to the minimal use of water, minimal costs for waste management, and no requirement to buy ice medium, the utility costs can be as low as about US\$4-5 per hour, depending on local electricity rates, local costs for producing compressed air, and other factors [46,47,96-98].

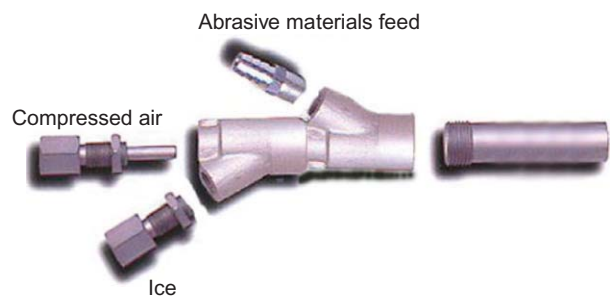


FIGURE 3.11 On-demand abrasive cleaning system with ice blasting [47].

Ice blasting is often compared with dry ice blasting for surface contaminant removal since both processes are similar in many aspects, such as the use of nonchemical phase change blasting media. Table 3.9 compares the costs of operations of both these processes for a stationary cleaning situation [46,47,96–101]. Clearly, water ice blasting is nearly three times less expensive than dry ice blasting with respect to operational costs. While there are regional variations and seasonal and special discounts, the costs shown in Table 3.9 are believed to be within 10%.

TABLE 3.9 Comparison of the Operational Costs of Water Ice Blasting and Dry Ice Blasting			
Item No.	Cost	Water Ice Blasting	Dry Ice Blasting
1	Production of dry ice pellets (pelletizer)	NA ^a	\$292,500
2	Storage and handling of dry ice pellets, 150 weeks × \$195	NA	\$29,250
3	Finance costs for equipment (3 years, 8%)	\$20,280	\$10,140
4	Depreciation of equipment	\$50,7000	\$25,350
5	Utility costs		
	● Electricity at \$0.078 per kWh	\$7020	\$1755
	● Electricity for CO ₂ ventilation	NA	\$7800
	● Costs for compressed air	\$46,800	\$46,800
	● Water	\$1170	NA
	● Recycling of waste water	\$910	NA
Operational costs for 6000 h usage in 3 years		\$126,880	\$413,595
Operational costs per hour		\$21.15	\$68.93
^a NA, not applicable.			

7 ADVANTAGES AND DISADVANTAGES

The advantages and disadvantages of ice blasting are as follows:

7.1 Advantages

1. Ice blasting is nonabrasive.
2. Ice does not leave a profile on the surface or damage the surface.
3. The temperature dependence of the hardness of ice makes it the only blast medium that can be adapted to the conditions and requirements of the process, from hard to very soft for application to delicate and highly sensitive substrates.
4. There are no negative thermal effects from the ice temperature.
5. Water ice at low temperature has nearly the same hardness as stainless steel or glass. It works like a hard abrasive medium to remove burrs and hard coatings.
6. No laborious postdeburring cleaning operations are required as compared to other deburring processes.
7. Low-temperature embrittlement of the material at cryogenic temperatures decreases the cutting force required during deburring operations.
8. There is no limit to the depth at which burrs can be removed.
9. It is safe and clean to work with, and it provides superior cleanliness.
10. The process is environmentally friendly. There are no CO₂ emissions.
11. The process is highly flexible with respect to the materials that can be decontaminated and the contaminants that can be removed.
12. Ice can be mixed with low amounts of an abrasive for enhanced cleaning effectiveness (abrasive on-demand system).
13. The process is amenable to a high degree of automation for in-line cleaning. It can also be configured as mobile technology for in-place cleaning of equipment.
14. Ice blasting is safe to use in confined areas or where adequate ventilation is an issue. No dust, fines, or fumes are generated or emitted.
15. The process has reliable and consistent operation. Ice making is a well-established industrial process.
16. Ice blasting is economical with low operating costs. It has low water consumption and uses normal tap water.
17. The process generates minimal waste, primarily water and removed contaminants. No additional waste is generated.

7.2 Disadvantages

1. Ice blasting is a line of sight process which limits efficient cleaning of parts with complex geometries.
2. The use of water induces corrosion or flash rusting on noncorrosion-resistant metals without drying, which makes subsequent paint or coating application more difficult and risky.

3. The process should not create a surface profile on a noncorrosion-resistant steel surface. A smooth surface profile is key parameter in superior paint performance.
4. The use of ice blasting on production equipment, including automated welding lines, presses, motors, and machine tools, can result in severe electrical problems or even burn up electrical motors.
5. It is specialized deburring technology and is a complementary process to hand deburring or water jet deburring.
6. No defined edge radius can be produced which makes it difficult to maintain high-precision profiles and shapes.
7. It requires the use of personnel safety equipment.

8 APPLICATIONS

There are many applications of ice blasting ranging from precision cleaning of semiconductor wafers and delicate items, such as books and antiques to removal of coatings and contaminants on a variety of substrates. Ice blasting has been effective in removing paint coatings from a variety of substrates without substrate damage, including delicate surfaces such as Kevlar and graphite-epoxy composites; however, the strip rate will vary greatly [47,102–104]. Conditions that affect the strip rate can include the blast pressure used, the thickness of the coating, substrate profile, adhesion strength, and age of the coating. Aircraft cleaning applications have included cleaning soiled aircraft coatings, removing decals from aircraft, and cleaning aircraft components such as gas turbine blades. Ice blasting has been successful in removing paints, grease, oil, grime, and other contaminants from a variety of glass surfaces, including windows, gauge panels, and controls. However, care must be exercised when blasting on these surfaces to avoid shattering. Typically, the blast pressure is turned down and a longer stand-off is required. Some examples of different ice blasting applications are described in the following sections.

8.1 Semiconductor Wafer Cleaning

A method has been developed for cleaning semiconductor wafers that employs fine and ultraclean ice particles between 0.1 and 300 μm diameter [105]. The impact on the wafer surface is controlled by adjusting the hardness of the ice particles, the jet pressure, and the jet angle. The ice particles act as a scrubber to precisely remove submicrometer particles ($<0.3 \mu\text{m}$ diameter) and organic contaminants such as fingerprints, grease, and oil-based inks, from a wafer surface. The cleaning method has been improved by adding a mechanism to accelerate the jet [106]. This is implemented by incorporating multiple ring-like electrodes, which are switched on by high-frequency AC power, and the length of each electrode in the direction of movement of the particles of ice is adjustable to the velocity of the ice particles. Another enhancement of the cleaning

method involves the use of ice particles containing hydrogen peroxide to clean the surface physically and chemically with the H_2O_2 solution obtained from the melting ice particles- H_2O_2 mixture [107].

Using the ice jet scrubbing method, several investigations have been carried out to successfully clean wafers and to evaluate the effect of major variables of the removal process on the cleaning efficiency [105,108–112]. Removal efficiency greater than 95% was obtained with ice particles. The removal efficiency is affected by the mass concentration of the particles, chamber pressure, and distance of the nozzle from the substrate.

8.2 Electronics and Photonics Applications

Ice blasting has been successfully used to remove contamination from a computer disk [113–116]. The disk surface was painted and then cleaned within 2 min at an ice flow rate of 35 g/min and ice particles of 0.5–3 mm diameter. After cleaning, the disk performed normally and no change in the sound of the audio recording on the disk was noticed. Other examples included cleaning of liquid crystal displays, circuit boards, solar panel elements, optical magnification lens, and emulsion removal from a photographic film surface [113–116]. No damage to the substrate was observed in visual or microscopic examination of the surfaces after cleaning.

In another study, the residual layer of highly adherent Wellbond[®] glue could be completely removed from the surface at a joint between a plastic and a rubber disk (10 cm diameter) as shown in Fig. 3.12 [117]. The cleaning duration was 2 min.

8.3 Automotive Components

Automated ice blasting systems have been demonstrated to be viable for various cleaning applications in the automotive industry, including cleaning of precision gears, aluminum valve bodies for transmissions, engine blocks, and aluminum cast parts [52,97,98]. In one gear cleaning application, the cleaning system could process 20 gears a second with ice particles sprayed at 0.45–0.52 MPa pressure. The system uses only about 76 L of water per hour of which about

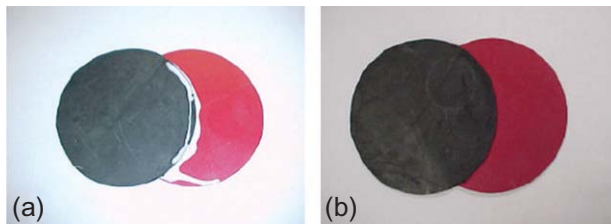


FIGURE 3.12 Removal of glue from plastic disk surfaces (a) before and (b) after ice blasting [117].

50% evaporates, leaving about 38 L that must be disposed. Over a three-month test period, more than 99% of the parts passed the first time through the cleaning process with obvious cost benefits.

8.4 Nuclear Applications

Ice blasting has replaced the traditional process of cleaning and degreasing the metal surfaces for bonding in the manufacture of Zircaloy bimetallic tubing for the nuclear fuel industry [46,118]. Surface contamination was evaluated by optically stimulated electron emission (OSEE). Figure 3.13 shows that the “as-received” sample has a very low OSEE value of 65 after machining, indicating the presence of machining oil and other contamination on the surface. The “conditioned” sample was scrubbed and thoroughly cleaned with methyl alcohol to establish an OSEE value of 999 as calibration [119]. Samples that are less clean will have lower OSEE values. One “conditioned” sample was subsequently cleaned by a standard acid etching process in HF-HNO₃, while the other sample was ice blasted. The acid-etched sample typically had an OSEE value of 375, suggesting that the acid etch process was contaminating rather than cleaning. The ice blasted sample gave the same OSEE value as the “conditioned” sample, confirming that ice blasting had the same degreasing capability as methyl alcohol with scrubbing.

In nuclear decontamination, INEEL (Idaho National Engineering and Environmental Laboratory) has evaluated both water ice blasting and dry ice (solid CO₂) for their effectiveness in removing surrogate contaminants Cs and Zr on stainless steel coupons [120]. Table 3.10 shows the relative decontamination recorded. At the optimal conditions, the SIMCON (*simulated contamination*) coupons were nearly completely decontaminated. While the results were not

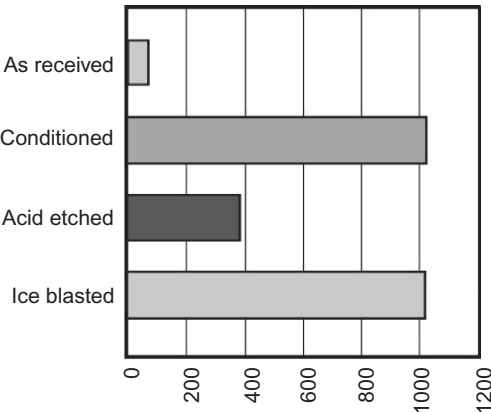


FIGURE 3.13 Comparison of OSEE values for Zircaloy bimetallic tubing after different cleaning processes [46,118].

TABLE 3.10 Decontamination Comparison of Water Ice and Dry Ice on Simulated Contamination Coupons [120]

	Cs (%)	Zr (%)
Dry ice blast	63	78
Ice blast	>98	>95

quite as high as more aggressive abrasive blasting techniques (which typically leave no contamination), there were significant advantages to using ice (melting to water) instead of an abrasive grit.

Ice blasting has been tested on radioactively contaminated lead bricks and contaminated tools and equipment [121,122]. Removal of fixed contamination from lead bricks was highly effective with decontamination factors (DFs)² ranging from >1.1 to >400, achieving free release levels without generating a mixed waste stream. Similarly, DFs ranging from >1.7 to >200 for β - γ radiation and >1.3 to >100 for α radiation were achieved for transferable contamination from tools and equipment. The process is less effective with contaminated tools and equipment that had chemically bonded contamination or contamination located in crevices in parts with complex geometries, since ice blasting does not abrade substrates and does not penetrate deeply into crevices.

Other examples of ice blasting include decontamination of a hot cell at the X-10 facility in Oak Ridge; decontaminating the containment building at the Oconee Reactor Unit 3; decontamination of tools and equipment at the Wolf Creek Reactor [96]; and final cleanup of the prototype vitrification facility in Marcoule, France [123,124]. It has been highly effective in decontaminating pump casings, valves, and primary circuit piping of a boiling water reactor, completely removing the soft crud on the inner surfaces and achieving a DF of 2 [125]. As an added benefit in maintenance, decontamination, and decommissioning activities in nuclear plants, ice blasting can lead to significant radiation dose reduction for workers. For example, during a maintenance outage at the Point Lepreau Generating Station in New Brunswick, Canada, gamma-radiation dose savings of nearly 1150 man-mSv was achieved by predecontamination of the steam generators by ice blasting before maintenance activities were performed [126]. General DFs of typically 5-6 were achieved with ice blasting. Nuclear criticality considerations must be addressed when dealing with special nuclear materials [127].

2. DF is a measure of the effectiveness of a decontamination process. It is the ratio of the original radioactivity to the remaining activity after decontamination.

8.5 Deburring of Components

An innovative application of ice blasting has been deburring and cleaning of components in the manufacturing process [48–50,94,128]. The process is most effective on soft metals, such as aluminum and brass, but it also can be used to deburr steel and iron, depending on burr size and how strongly the burr is attached to the part. It has also been successfully applied on nonmetallic materials such as polyethylene and polypropylene, although the process parameters must be optimized for the application [50]. Complex components can be easily deburred (Fig. 3.14), and large and small burrs can be removed (Fig. 3.15). Small burrs are generally difficult to remove because they have small frontal surface areas and long-hinged sections. With ice blasting, the parts are cleaned as well as deburred, leaving a clean and smooth surface. For most metal components, the parts must be dried immediately following the deburring step to prevent corrosion, oxidation, and flash rusting.

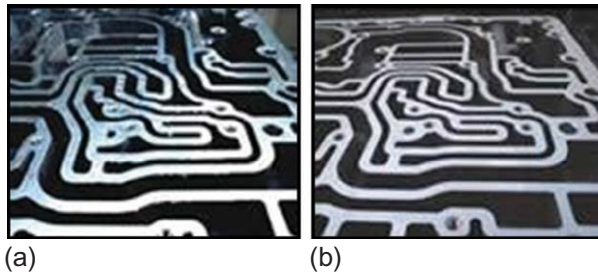


FIGURE 3.14 Deburring of a complex transmission valve body component using ice blasting: (a) before deburring and (b) after deburring [94].

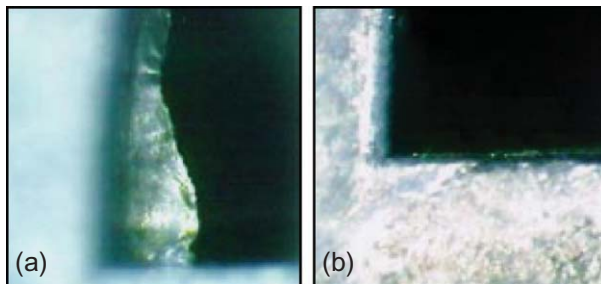


FIGURE 3.15 Removal of a large burr on a zinc diecast valve housing component with ice blasting: (a) before deburring and (b) after deburring [128].

8.6 Recycling and Recovery

Several recycling and recovery applications have been demonstrated with ice blasting. It has been used for removing misaligned insulation foam from defective parts so that new foam can be applied, thereby salvaging the higher value component. The cost of this ice blasting recovery operation is about US\$0.35 per piece [47]. One manufacturer in the United States produces about 9 million kilograms of scrap vinyl from dashboard and interior components of automobiles. Ice blasting was used to remove the glue and foam from scrap vinyl to below 2% contamination at a cost of between US\$1.60 and US\$2.30 per kilogram. Although the total cost of disposal at a landfill is about US\$1.10 per kilogram, disposal essentially results in the loss of recyclable noncontaminated vinyl with a commercial value of about US\$3.00 per kilogram [47].

8.7 Facility Decommissioning

Ice blasting was used to clean the concrete floor, saturated with shot, oil and grime, of a manufacturing plant for shot blasting and heat treatment prior to its closing [47]. Ice blasting was first used to remove the oil-saturated shot from the floor. The cleaning rate was about 25 m²/h. A biodegradable detergent was sprayed on the concrete floor to extract the oil. The floor was ice blasted again to produce a floor meeting free release specification. The production rate was approximately 40 m²/h. Total liquid waste generated averaged 6.4 L/100 m² for this double application of ice blasting [47].

Waste minimization and superior cleanliness combined to position ice blasting as a viable process for structural rehabilitation or decommissioning of facilities. Its effectiveness enabled a contractor to meet production in the asbestos abatement of the 43rd floor of a high-rise building in Portland, Oregon, and lead-based paint abatement on two bridges by the New York Department of Transportation.

The U.S. Navy has used ice blasting to remove loose lead paint from buildings prior to decommissioning or renovation for alternative use. There was no liquid waste generated, and the airborne lead dust level was significantly below specification. The effectiveness of the process produced tremendous savings compared to manual cleaning as shown in Table 3.11 for one building at the Washington Navy Yard [129].

8.8 Restoration of Historic Building

The Aberdeen Pavilion in Ottawa, Canada, is a historic structure made entirely of metal [130]. Ice blasting has been employed for effective surface preparation of different parts of the building covered with loose and bound rust, paint, embedded salt, and layers of lead-based coating. The coated materials were

TABLE 3.11 Comparison of Ice Blasting Process Versus Manual Process for Lead Paint Abatement in One Building in the Washington Navy Yard [129]

Parameter	Manual Cleaning	Ice Blasting
Process time	900 h	375 h
Personnel	8-10 men	5 men
Equipment	4 high lifts	2 high lifts
Cost	\$432,000	\$180,000

removed without removing any of underlying original substrate material. The cleaning rate was approximately 9 m²/h on the flat roof section and 5-7 m²/h on the corrugated section.

8.9 Oil Field Equipment Cleaning

Ice blasting has been used for cleaning in an oil field environment [47,131]. Examples of equipment cleaned included an oil storage tank, a rod pumping unit, a road grader, and a wellhead. The original paint of the oil storage tank, a light beige color, was completely coated with a thin layer unrefined 30°API sour crude oil. The pumping unit was heavily overlaid with the same sour crude oil. The grader was covered with a mixture of spent hydraulic fluid and dirt. The wellhead was coated with a mixture of crude oil, paraffin and dirt, hardened over the years by heat from steam flooding operations. Two panels of the tank, each measuring about 1.2 m × 1.5 m, were cleaned in approximately 20 min. Although the cleaned surface had been permanently stained by the oil spill, ice blasting removed all surface oil and left the surface suitable for repainting. The results of cleaning of the pumping unit covered with the crude oil were even more striking. The unit was cleaned beginning at the base, proceeding up the support legs, and ending with the horse head. There was no noticeable pooling or collecting of water. Oil was quickly removed over a 30-min period. At the end of the cleaning process, it was observed that the unit was nearly spotless, there was no oily residue and a minimal amount of waste had collected around the base of the unit. No paint was removed during the cleaning procedure. Subsequent cleaning efforts on the road grader and the wellhead using the ice blasting system produced similar positive results. A minimal amount of waste moisture was generated, thereby reducing cleanup and disposal costs.

Ice blasting has been very effective in cleaning crude oil from burners in a furnace, removing insulation from pipes, degreasing casings and drill pipes, cleaning spilled antioxidants, and bitumen from aluminum moldings (Fig. 3.16) [47].

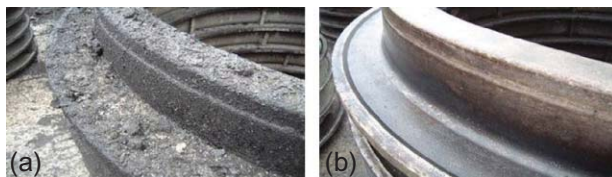


FIGURE 3.16 Cleaning aluminum moldings contaminated with bitumen by ice blasting: (a) before cleaning and (b) after cleaning [47].

8.10 Cleaning Pipes

Although strictly not a blasting technique, ice pigging is an innovative method that utilizes ice-water slurry to remove contamination from pipes [132–134]. The underlying concept is that when an ice slurry, typically between 50% and 90% ice fraction, is propelled through pipes at modest speeds, the wall shear is 2–4 orders of magnitude higher than with water alone traveling in the pipe at the same speed. Thus, even with relatively low speeds, the “ice pig” is able to achieve efficient cleaning and removal of loose materials, without damaging the pipe walls. The high ice fraction allows the ice pig to hold together under continuous shear. This technology has the advantage that the ice pig changes its shape to fit the containing topology; hence, it is able to navigate bends, contraction/expansion geometries, and partly open valves, while cleaning the containment walls and transporting particulates. Lastly, the ice pig is guaranteed never to get stuck, as it will simply melt away, if left for sufficient time. The technique is being used worldwide in the water industry, food processing, pharmaceutical manufacturing, chemical production, surface coatings industry, and hydrocarbon recovery.

9 SUMMARY AND CONCLUSIONS

Ice blasting is a simple, nonabrasive, cleaning process that uses ice crystals as impact medium for removing surface contaminants without the use of chemicals, abrasive materials, high temperatures, or steam. The technology employs ordinary tap water, compressed air and electricity to create an environmentally friendly, cost-effective method for surface cleaning. Ice as a phase change blast medium has the ability to change its physical state to liquid water which flushes contaminants from the surface, leaving no solid residue after blasting. The technique can be used for cleaning surfaces, removing paint, or stripping contaminants from a surface. It can also be used to remove loose material, blips, and burrs from metal components after machining, and even softer materials, such as organic polymeric materials, including plastic and rubber components, can be processed. Compared to other blast cleaning processes, ice blasting does not accumulate ice particles in the waste and thus significantly reduces secondary waste. Applications of ice blasting range from precision cleaning of

semiconductor wafers and delicate items, such as books and antiques, to removal of contaminants on a variety of substrates. Ice blasting has been effective in removing paint coatings from a variety of substrates without substrate damage, including delicate surfaces such as Kevlar and graphite-epoxy composites, as well as removing paints, grease, oil, grime, and other contaminants from a variety of glass surfaces, including windows, gauge panels, and controls.

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DISCLAIMER

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